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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.4

$f(x) = x^6 + x^3 - x^2 + 4$ in het punt $a = 1$

$f(1) = 1 + 1 - 1 + 4 = 5$

$f'(x) = 6x^5 + 3x^2 - 2x$

$f'(1) = 6 + 3 - 2 = 7$

$f''(x) = 30x^4 + 6x - 2$

$f''(1) = 30 + 6 - 2 = 34$

$f^{(3)}(x) = 120x^3 + 6$

$f^{(3)}(1) = 120 + 6 = 126$

$f^{(4)}(x) = 360x^2$

$f^{(4)}(1) = 360$

$f^{(5)}(x) = 720x$

$f^{(5)}(1) = 720$

$f^{(6)}(x) = 720$

$f^{(6)}(1) = 720$

$$T_6(f,1)(x) = 5 + \frac{7}{1}(x-1) + \frac{34}{2}(x-1)^2 + \frac{126}{6}(x-1)^3 + \frac{360}{24}(x-1)^4$$
$$+\frac{720}{120}(x-1)^5 + \frac{720}{720}(x-1)^6 = 5 + 7(x-1) + 17(x-1)^2 + 21(x-1)^3$$
$$+15(x-1)^4 + 6(x-1)^5 + (x-1)^6$$

References